Sreya Vadapalli

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Professional Summary

Biomedical and public health data scientist with 3+ years of experience applying machine learning , artificial intelligence, and statistical modeling across epidemiology, precision medicine, and behavioral health research.

Professional Experience

Rogers Behavioral Health, Milwaukee **Aug 2023 - June 2026**

Research Associate II

- Quantified a $4.2B annual healthcare burden across 108 rare genetic disorders with primary neuropsychiatric symptoms and a 7.7-year diagnostic delay, building the case for earlier genetic testing.
- Analyzed clinical outcomes (quality of life, depression, anxiety, OCD severity) and medication utilization (polypharmacy) across matched cohorts at admission and discharge in retrospective studies of OCD and autism spectrum disorder.
- Evaluated a quality improvement initiative on anxiolytic drug management in eating disorder patients receiving exposure-based CBT.
- Built an integrated data pipeline to process, extract, and investigate patient data.

Rutgers University, New Jersey (*during MBS degree)*

Robert Wood Johnson Medical School, NJ **Aug 2022 - May 2023**

Research Assistant

- Implemented ensemble decision forests model on EEG data to study post-ictogenesis achieving **91.4%** accuracy and **0.97** AUC.
- Performed neural networks for auto-segmentation of pre-ictal and ictal seizure phases in epilepsy.
- Analyzed Racine score associations with the number of seizures and days using statistical methodologies.

Rutgers MBS Externship Exchange, NJ **Sep 2022 - Dec 2022**

Data Science Extern (COVID-19 Epidemiology Bootcamp)

- Assembled and curated multi-source COVID-19 data (cases, fatalities, NWSS wastewater, county/state) and developed predictive ML models (neural networks, logistic regression, and XGBoost), achieving **77%** accuracy in state-level variant prediction with XGBoost.

Rutgers Institute of Health, Health care Policy, and Aging Research, NJ **Nov 2021 - May 2022**

Data Science Research Assistant and Teaching Assistant

- Reviewed and analyzed **32** AI, ML approaches applied in whole genome, whole exome sequencing and gene expression data, identified widely adopted predictive algorithms for precision medicine across multiple diseases.
- Contributed to RNA-seq driven transcriptomic analysis to identify **3** highly expressed AF-associated genes (*PF4, PPBP, MYL4*) and their phenotypic associations across race and gender.
- Led weekly discussions and held office hours for 20 undergraduate students enrolled in Byrne Seminar - Practicing Precision Medicine with Data Analysis.

Nizam's Institute of Medical Sciences

Medical Biotech and Statistics Intern **May 2019 - Jul 2019**

- Investigated the association between Taq1A and Taq1B genetic polymorphisms and Parkinson's disease susceptibility in South Indian populations.
- Compared genotype and allele frequency distributions between Parkison's disease cases and controls to identify statistically significant genetic differences.

Publications

1. Artificial intelligence & machine learning approaches using gene expression and variant data for personalized medicine.(Oxford, Briefings in Bioinformatics)
2. Artificial Intelligence, Healthcare, Clinical-Genomics, and Pharmacogenomics Approaches in Precision Medicine. (Frontiers, Genetics)
3. RNA-seq-driven expression analysis to investigate cardiovascular disease genes with associated phenotypes among Atrial Fibrillation patients. (Wiley, Clinical and Translational Medicine)
4. Polypharmacy and pharmacogenomics in high-acuity behavioral health care for autism spectrum disorder: a retrospective study. (Springer, CAPMH)
5. Pharmacogenomic-guided prescribing and polypharmacy across age groups in obsessive-compulsive disorder: a retrospective study. (Springer, BMC)
6. Bridging psychiatry and rare genetic diseases: a scoping review of therapeutic strategies and diagnostic delay paired with healthcare economic burden analysis. (Springer, Orphanet Journal of Rare Diseases)
7. Multi-Frequency Interpolation X-talk Removal Algorithm: Enabling Combinations of Concurrent Optogenetics and Lock-in Amplification Fiber Photometry via Removal of Optogenetic Stimulation Crosstalk. (ACS, Chemical Neuroscience)
8. P342: Utilization of pharmacogenomics in a child and adolescent autism spectrum disorder depression and anxiety program: A retrospective study (Abstract, ACMG, Genetics in Medicine)
9. 3.38 Utilization of Pharmacogenomics for ASD: A Retrospective Study (Journal of American Academy of Child and Adolescent Psychiatry)
10. Anxiolytic management in eating disorder patients receiving cognitive behavioral therapy: A quality improvement brief report. (medRxiv, Preprint)

Presentations and Talks

- Presented a poster titled Tailoring Exposure and Response Prevention to Treat Disgust OCD at the International OCD Foundation conference in Chicago
- Invited talk / Webinar : Clean Sheets or Muddy Boots: A Discussion of Curated and Field Data Sets (Midwest Big Data Innovation Hub Data Science Student Groups Webinar)

Programming: Python (pandas, NumPy, scikit-learn, PyTorch), R, SQL, Tableau, MATLAB
Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Deep Learning
Gen AI: LLMs, NLP , Transformers, RAG
Genomics: NGS Data Processing,WES, Variant Calling (GATK), Alignment (BWA)

Academic Projects

LitLens

- Built a biomedical RAG app using React, FastAPI, Claude API, SPECTER embeddings, and ChromaDB to parse, retrieve, and summarize research papers.
- Designed an evaluation framework for retrieval and generation quality using a hand-labeled gold set, including Hit@k, MRR, and claim-grounding checks.
- Improved retrieval precision through source-scoped filtering, raising Hit@5 from 0.55 to 0.75.

Life Expectancy Prediction

- Fitted a Multiple Linear Regression model predicting life expectancy on WHO data (2000-2015), Ridge/Lasso regularization and country fixed effects improved R^2 from 0.81 to 0.94.

Exploratory Analysis of Poverty in NYC

- Supervised ML implementation to predict poverty in the NYC population resulting in an accuracy of 84% by applying and comparing SMOTE, random oversampling, and random under-sampling to address class imbalance.

Leadership and Achievements

- Chief Head, President, and Editorial in charge of Bioengineering and Biotechnology Club of C.B.I.T, 2019-2020.
- Presented a chapter presentation on projects done in C.B.I.T at The All-India EWB CHAPTERS MEET, Chennai, 2018.
- Head of Logistics, Model United Nation CBIT 2018, Deputy Head of Logistics, Model United Nation CBIT, 2017.
- Core-Committee Head - TEDx CBIT, 2018.

Education

Rutgers University (2023) New Brunswick, New Jersey.
Master of Business and Science in Analytics, GPA: 3.7
Awards: MBS fellowship Aug 2022 & Jan 2023

C.B.I.T, Osmania University (2020) Hyderabad, India.
Bachelor of Technology, Biotechnology, GPA: 8.41/10 (3.9)
Awards: Silver Medal for Academic Excellence